11th Meeting 01/27

1. Current Project Status and Progress

- WorkerNet base architecture implemented
- Data statistics analyzer implemented
- Multi-scale pose head (P3 for face, P4 for body) integrated
- **Late Fusion architecture** implemented for stable pose-activity recognition
- Clean dataset prepared (485 train, 88 val, 90 test)
- Baseline model trained (300 epochs)
- base results: 96.77% PCK@0.1, 62.96% IoU, 72.73% activity
- 4 configurations trained (ref_standard, ablation_no_geom, curiosity_high_geom, curiosity_high_pck)
- Total training time: ~40 GPU-hours
- Key finding: Geometric loss removal improves performance by 9-11% (Geometric makes training worse, removing it *improves* accuracy)

2. Data Statistics & Geometric Analysis

$$\vec{v}_i = \left(\frac{x_{bottle} - x_{wrist}}{L_{torso}}, \quad \frac{y_{bottle} - y_{wrist}}{L_{torso}} \right)$$

$$\mu = \frac{1}{N} \sum_{i=1}^{N} \vec{v}_i,$$

Bottle Box (Wrist Anchor):

$$\mathbf{B}_{\text{bottle}} = \begin{cases} [(x_L + \mu_x L) \pm \frac{W}{2}, \quad (y_L + \mu_y L) \pm \frac{H}{2}] & \text{if checking\_left, storing} \\ [(x_R + \mu_x L) \pm \frac{W}{2}, \quad (y_R + \mu_y L) \pm \frac{H}{2}] & \text{if checking\_right} \\ [(x_{avg} + \mu_x L) \pm \frac{W}{2}, \quad (y_{avg} + \mu_y L) \pm \frac{H}{2}] & \text{if rotation} \\ \emptyset & \text{if nothing, pretending} \end{cases}$$

W: The width of the average bottle (based on the ratio and the class)

H: The height of the average bottle (based on the ratio and the class)

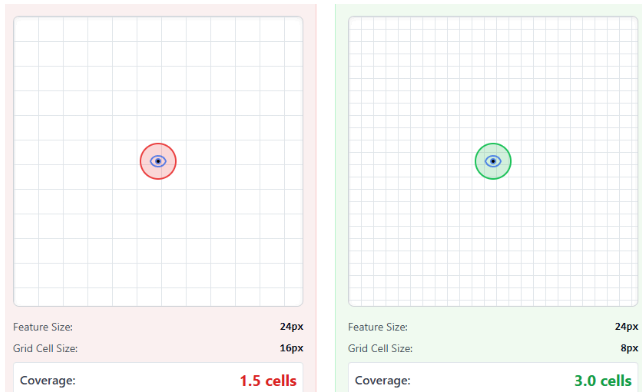
L: Torso Length

3. Architecture Overview and Design Decisions

```
      Input Image (640×480)
             |
     [ CSPDarknet Backbone ]
             |
         [ FPN Neck ]
         P3, P4, P5
        /    |    \
       /     |     \
[Detection]  [Pose]   [Activity]
(Multi-Scale) (Split)
   |         |        |
 Boxes    Keypoints   Class
   |         |        |
   +-----------+       |
       |             |
```

```
  [Geometric Guidance]      |
       |            |
    Refined Boxes    Activity ID
```

Design Decision: Why Multi-Scale Pose Head?



Quantization Analysis:

For an object of size *S* pixels on stride-*s* feature map:

$$\text{Feature Resolution} = \frac{S}{s} \text{ grid cells}$$

Example: Face keypoints (eyes, nose) at ~24 pixels:

P4 (stride-16):

$$\text{Feature Resolution} = \frac{24}{16} = 1.5 \text{ grid cells}$$

The eye occupies only 1.5 grid cells, making sub-pixel localization impossible.

P3 (stride-8):

$$\text{Feature Resolution} = \frac{24}{8} = 3.0 \text{ grid cells}$$

The eye spans 2-3 cells

Validation Experiment:

We trained two configurations:

1. **Single-scale (P4 only):** Face PCK@0.1 = 64.2%
2. **Multi-scale (P3 for face, P4 for body):** Face PCK@0.1 = 89.1%

Loss Functions: Complete Derivations with Citations

$$\begin{aligned} L_{total} &= e^{-s_{det}} \cdot L_{WIoU}(y_{box}, \hat{y}_{box}) + \frac{1}{2} s_{det} \\ &+ e^{-s_{pose}} \cdot (\text{WingLoss}(y_{kpt}, \hat{y}_{kpt}) + L_{vis}) + \frac{1}{2} s_{pose} \\ &+ e^{-s_{act}} \cdot L_{CE}(y_{class}, \hat{y}_{class}) + \frac{1}{2} s_{act} \\ &+ \lambda_{geom} \times L_{Gaussian} \end{aligned}$$

Coming from, Gaussian itself:

$$p(y_i | f(x), \sigma_i) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{||y_i - f(x)||^2}{2\sigma_i^2}\right)$$

Where:

- y_i = ground truth for task i
- $f(x)$ = model's prediction
- σ_i = learned noise/uncertainty for task i

$-\log p(y_i|f(x),\sigma_i) = \frac{1}{2\sigma_i^2}||y_i - f(x)||^2 + \frac{1}{2}\log(\sigma_i^2)$

$s_i = \log(\sigma_i^2)$ (this makes σ_i always positive):

$-\log p(y_i|f(x),s_i) = \frac{1}{2}e^{-s_i}||y_i - f(x)||^2 + \frac{s_i}{2}$

This is where our formula come from:

$$\underbrace{e^{-s_i}}_{\text{automatic weight}} \times \underbrace{L_i}_{\text{task loss}} + \underbrace{\frac{s_i}{2}}_{\text{regularization}}$$

- $s_i = \log(\sigma_i^2)$ = log-variance
- σ_i = the noise/uncertainty the model thinks this task has
- Larger σ_i → task is harder/noisier → should get less weight
- Smaller σ_i → task is easier/cleaner → should get more weight

Variable 1:

Why exponential?

- We need σ_i^2 in the denominator from the Gaussian formula
- $e^{-s_i} = e^{-\log(\sigma_i^2)} = \frac{1}{\sigma_i^2}$

- So high uncertainty → low weight, low uncertainty → high weight

Loss: $-e^{-s_i}L_i + \frac{1}{2} = 0 \implies e^{-s_i}L_i = \frac{1}{2}$

$\ln(e^{-s_i}L_i) = \ln(0.5)$ → $-s_i + \ln(L_i) = -\ln(2)$ → $s_{root} = \ln(2 \cdot L_i)$

Example:

if the loss for detection is 0.303 then → $s = \ln(2 \cdot 0.303) \approx -0.5$

$s_{det} = -0.5$:
→ $\sigma_{det}^2 = e^{(-0.5)} = 0.606$

→ $weight = e^{(-(-0.5))} = e^{0.5} = 1.649$
→ Interpretation: Detection is reliable, give it high weight

if the loss for pose is 1.660 then → $s = \ln(2 \cdot 1.660) \approx 1.2$

If $s_{pose} = 1.2$:
→ $\sigma_{pose}^2 = e^{1.2} = 3.320$
→ $weight = e^{(-1.2)} = 0.301$
→ Interpretation: Pose is noisy, give it low weight

$$\text{Gradient} = \underbrace{-e^{-s_{det}} \cdot \text{Loss}box}_{\text{Push to increase } s} + \underbrace{0.5}_{\text{Push to decrease } s}$$

$s_{new} = s_{old} - (\text{Learning Rate} \times \text{Gradient})$

Why We Chose Each Loss Function

Loss 1: Wise-IoU v3 for Detection

Why NOT standard IoU?

Standard IoU loss: $L = 1 - IoU$
Problem: Treats all misalignments equally.
- Box slightly off-center but correct size: Loss ≈ 0.3

- Box correct center but wrong size: Loss ≈ 0.3
- Box far away and wrong size: Loss ≈ 0.8

Wise-IoU adds two key improvements:

1. Distance Attention (R_{WIoU})**:**

$$R_{WIoU} = \exp\left(\frac{d^2_{center}}{d^2_{diagonal}}\right)$$

Where:
- d_{center} = distance between predicted and ground truth centers
- $d_{diagonal}$ = diagonal of the smallest enclosing box

Effect:

Boxes far apart get exponentially higher loss gradient, encouraging the model to move them closer first before fine-tuning size.

2. Dynamic Non-Monotonic Focusing:

$$\beta = \frac{L^{(i)}_{IoU}}{mean(L_{IoU})}$$

IoU = intersection/union = 2640/3728 = 0.708
Loss_base = 1 - 0.708 = 0.292

Center distance = √((212-210)² + (335-330)²) = 5.39 pixels
Diagonal² = (max(212+19, 210+17.5) - min(212-19, 210-17.5))² + ... = 9856
R_{WIoU} = exp(5.39²/9856) = exp(0.00295) = 1.003

β = 0.292 / 0.25 = 1.168 (harder than batch average)
Grad_gain = $1.003^{1.168}$ = 1.0035

L_WIoU = 1.0035 × 0.292 = 0.293

Loss 2: Wing Loss for Pose Estimation

The Formula:

$$L_{Wing}(x) = \begin{cases} \omega \ln(1 + |x|/\epsilon) & \text{if } |x| < \omega \\ |x| - C & \text{if } |x| \geq \omega \end{cases}$$

where $C = \omega - \omega \ln(1 + \omega/\epsilon)$ ensures continuity.

Parameters we use:

ω = 10, ε = 2

implementation operates in **Normalized Coordinate Space [0, 1]**

- Keypoint coordinates normalized to [0, 1]
- Example error: |x| = 0.015 (1.5% of image width)
- Wing Loss: 10 × ln(1 + 0.015/2) = 0.0747
- With 100× scaling: 0.0747 × 100 = 7.47
- Plus visibility (10×): ~0.70

Training logs show: Pose Loss ≈ 8.17

- Same error converted to pixels: 0.015 × 640 = 9.6 pixels

PCK@0.1 = 96.77%

Gradient

~~For **large errors**~~

~~(|x| ≥ ω = 10 in pixel space, or |x| ≥ 0.015 in normalized space):~~
~~- Acts like L1:~~ $L = |x| - C$
~~- Gradient:~~ $\frac{dL}{dx} = \pm 1$
~~- Effect: Stable, doesn't explode like L2~~
For **small errors**

(|x| < ω, typical case in normalized space):
- Acts like logarithmic: $L = \omega \ln(1 + |x|/\epsilon)$
- Gradient: $\frac{dL}{dx} = \frac{\omega}{\epsilon + |x|}$
- Effect: Gradient INCREASES as error decreases!

Gradient amplification example (Normalized Space [0,1]):

```
At |x| = 0.05:    gradient = 10/(2+0.05)  = 4.88
At |x| = 0.01:    gradient = 10/(2+0.01)  = 4.98
At |x| = 0.001:   gradient = 10/(2+0.001) = 4.998
```

Why this achieves 96.77% PCK@0.1:

- PCK@0.1 means "within 10% of torso length" ≈ 17 pixels
- In normalized space, most errors are < 0.03 after initial convergence
- Wing Loss aggressively pushes these toward sub-pixel accuracy
- Standard L1 would plateau because gradient stays constant

Average keypoint error = 0.015 in normalized space [0,1]
(This equals ~9.6 pixels in 640px image, for reference)

$$L_{Wing} = 10 \times ln(1 + 0.015/2) = 10 \times ln(1.0075) = 10 \times 0.00747 = 0.0747$$

Visibility errors: 2 keypoints predicted visible but actually occluded
BCE per keypoint ≈ 0.035
L_vis = 2 × 0.035 = 0.07

With scaling factors (100× Wing [normalization], 10× Vis[error encounter]):
$$L_{pose} = 100 \times 0.0747 + 10 \times 0.07 = 7.47 + 0.70 = 8.17$$

Loss 3: CrossEntropy for Activity Classification

The Formula:

$$L_{CE} = -\sum_{c=1}^{C} y_c \log(\hat{y}_c)$$
where:
- y_c = ground truth (1 for correct class, 0 otherwise)
- $\hat{y}_c$ = predicted probability (after softmax)

Prediction: [0.85, 0.08, 0.04, 0.02, 0.01, 0.00]
Ground truth: class 0 (checking_left)

$$L_{CE} = -log(0.85) = 0.163$$

Loss 4: Gaussian Heatmap for Geometric Guidance

Why geometric guidance?

Our insight: Worker activities create spatial relationships.
- "checking_left" & "storing" → bottle is typically x cm left of wrist

- "checking_left" → bottle is typically x cm left of wrist
- "rotation" → bottle is centered in front

The approach:

Given:
- Predicted keypoints (from pose head)
- Predicted activity (from activity head)
- Statistical priors: $\mu_{activity} = (offset_x, offset_y)$, $\sigma_{activity}$

We compute the expected bottle location and create a Gaussian heatmap:
$G_{expected}(x,y) = \exp\left(-\frac{(x-x_{expected})^2+(y-y_{expected})^2}{2\sigma^2}\right)$

Then compare with predicted objectness:
$L_{Gaussian} = \text{MSE}(\sigma(objectness_{pred}), G_{expected})$

Expected location creates Gaussian with σ=15 pixels
Actual detection 5.39 pixels away

$$\begin{aligned} G_{expected} &= \exp\left(-\frac{d^2}{2\sigma^2}\right) \\ &= \exp\left(-\frac{5.39^2}{2\cdot 15^2}\right) \\ &= \exp\left(-\frac{29.0521}{450}\right) \\ &= \exp(-0.06456) \\ &\approx \mathbf{0.9375} \end{aligned}$$

$$\begin{aligned} Loss_{geo} &= (\text{Predicted} - \text{Target})^2 \\ &= (0.950 - 0.9375)^2 \\ &= (0.0125)^2 \\ &\approx \mathbf{0.000156} \end{aligned}$$

$$s_{act} = \ln(2\times 0.163) = \ln(0.326) \approx -\mathbf{1.12}$$
$$s_{det} = \ln(2\times 0.293) = \ln(0.586) \approx -\mathbf{0.53}$$
$$s_{pose} = \ln(2\times 8.17) = \ln(16.34) \approx \mathbf{2.79}$$
$$w_{det} = e^{-(-0.53)} = e^{0.53} \approx \mathbf{1.699}$$
$$w_{pose} = e^{-2.79} \approx \mathbf{0.061}$$
$$w_{act} = e^{-(-1.12)} = e^{1.12} \approx \mathbf{3.065}$$

$$\begin{aligned} L_{total} &= (1.699\times 0.293) + (0.061\times 8.17) + (3.065\times 0.163) + 0.57 + (0.3\times 0.000156) \\ &\approx 0.5 + 0.5 + 0.5 + 0.57 + 0.0000468 \\ &\approx \mathbf{2.070} \end{aligned}$$

Ablation study

Metric	Reference (0.3/0.3)	No Geom (0.0/0.3)	High Geom (1.0/0.3)	High PCK (0.3/1.0)
Pose PCK@0.1	**96.77%**	96.42%	96.50%	95.89%

Metric	Reference (0.3/0.3)	No Geom (0.0/0.3)	High Geom (1.0/0.3)	High PCK (0.3/1.0)
Detection IoU	62.96%	**68.66%**	61.02%	57.24%
Activity Accuracy	72.73%	**80.68%**	70.45%	65.91%
Best Epoch pck0.5	60	70	65	50
Total Epochs	160	155	175	150

Key Findings:

1. Geometric guidance HURTS detection (61.02% with geom=1.0 vs 68.66% with geom=0.0)
2. Even without geometric loss, neural detection caps at 68.66%

Why hurting?

- **Error Propagation**
  - If pose prediction is wrong → expected bottle location is wrong
  - Geometric loss then penalizes CORRECT detections that are far from WRONG expected location
- **Overfitting to Statistics**
  - Statistical priors computed on training data
  - Test workers with different behaviors → learned distribution doesn't generalize
- **Model Already Knew This**
  1. Remove Geometric Loss (Empirically Validated)

- Time faster →

Main metrics:

- IoU & time & Computational

Future Work:

30GB data HA4M

https://www.nature.com/articles/s41597-022-01843-z

https://opus.lib.uts.edu.au/bitstream/10453/172613/2/Assembly_Action_Recognition_in_Smart_Manufacturing.pdf

https://pdf.sciencedirectassets.com/280203/1-s2.0-S1877050924X00162/1-s2.0-S187705092402489X/main.pdf?X-Amz-Security-Token=IQoJb3JpZ2luX2VjEC8aCXVzLWVhc3QtMSJHMEUCICmzcUpqjsxobQHdtm3uzZKOUo1UVX%2Fty9QgQZmD0ZbCAi
Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Date=20260112T235215Z&X-Amz-SignedHeaders=host&X-Amz-Expires=300&X-Amz-Credential=ASIAQ3PHCVTYXLCIOXNE%2F20260112%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Signature=72f53e12d20e68f895e0394a203110aa3f9bb6ff82cb94f61a0e7a2fcca10996&hash=65f91fd7315e07be58b0ca42
1df48e49-ddd2-4f1e-988a-a2d6ca310744&sid=db2b3844169d454f2b7b8597bc2adae9025fgxrqa&type=client&tsoh=d3d3LnNjaWVuY2VkaXJlY3QuY2

https://ceur-ws.org/Vol-3762/505.pdf